

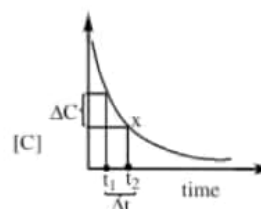
2. For the reaction $A \rightarrow B$; following curves represent conc. vs time



The correct curves are

- 1) 1, 2 only 2) 2, 3 only 3) 1, 4 only 4) 3, 4 only
3. For $N_2 + 3H_2 \rightarrow 2NH_3$, rates of disappearance of N_2 and H_2 and rate of appearance of NH_3 respectively are a, b and c, then
- 1) $a > b > c$ 2) $a < c < b$ 3) $a = b > c$ 4) $a = b = c$
4. The reaction $CH_3COOC_2H_5 + H_2O \xrightarrow{H^+}$ products is a/an _____ process
- A) Instantaneous B) Spontaneous C) Moderately slow
- Then correct statement(s) is/are
- 1) A & C 2) B & C 3) A only 4) C only

5. From the graph the value $\frac{\Delta c}{\Delta t}$ and the value of rate of reaction at X respectively are called

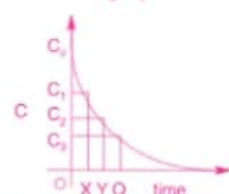


- 1) Average rate and instantaneous rate
2) Instantaneous rate and average rate
3) Average rate only
4) Instantaneous rate only
6. In the following reaction, how is the rate of appearance of the underlined product related to the rate of disappearance of the underlined reactant?



- 1) $\frac{d[Br_2]}{dt} = -\frac{5}{3} \frac{d[Br^-]}{dt}$ 2) $\frac{d[Br_2]}{dt} = -\frac{d[Br^-]}{dt}$ 3) $\frac{d[Br_2]}{dt} = -\frac{d[Br^-]}{dt}$ 4) $\frac{d[Br_2]}{dt} = -\frac{3}{5} \frac{d[Br^-]}{dt}$

7. From the graph.



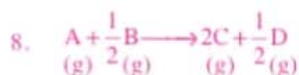
X = 10 min;

Y = 20 min;

Q = 30 min

Now, correct relationship is

- 1) $\frac{C_0 - C_1}{X} = \frac{C_1 - C_2}{Y} = \frac{C_2 - C_3}{Q}$ 2) $\frac{C_2 - C_3}{Q} > \frac{C_1 - C_2}{Y} > \frac{C_0 - C_1}{X}$
- 3) $\frac{C_0 - C_1}{X} > \frac{C_1 - C_2}{Y - X} > \frac{C_2 - C_3}{Q - Y}$ 4) $\frac{C_1 - C_3}{Q - X} = \frac{C_0 - C_2}{Y}$



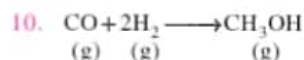
The rate of disappearance of 'B' is 'x'. What is the rate of appearance of 'C' ?

- 1) 3x 2) x/2 3) 2x 4) 4x

9. In which of the following cases, rate of disappearance of any reactant at a given instant equals to rate of appearance of any product



Numerical Value Type Questions



In the above reaction Hydrogen disappears at the rate of 0.2 gm/sec. What is the rate of appearance of methanol (gm/sec) at that moment?

11. In the reaction $2A + B \rightarrow 2C + D$, 5 mol/lit of 'A' are allowed to react with 3 mol/lit of B. After 5 seconds the concentration of 'A' was found to be 4 molar. What is the rate of reaction in terms of A?

12. For $3A \rightarrow xB$, $\frac{d[B]}{dt}$ is found to be $2/3^{rd}$ of $\frac{d[A]}{dt}$. Then, the value of x is

LEVEL-II (ADVANCED)

Straight Objective Type Questions

1. For the reaction $N_2O_5 \rightarrow 2NO_2 + \frac{1}{2}O_2$; Given $-\frac{d[N_2O_5]}{dt} = K_1[N_2O_5]$, $\frac{d[NO_2]}{dt} = K_2[N_2O_5]$ and $\frac{d[O_2]}{dt} = K_3[N_2O_5]$. The relation in between K_1 , K_2 and K_3 is :
 a) $2K_1 = K_2 = 4K_3$ b) $K_1 = K_2 = K_3$ c) $2K_1 = 4K_2 = K_3$ d) None of these
2. Hydrogenation of vegetable ghee at $25^\circ C$ reduces pressure of H_2 from 2 atm to 1.2 atm in 50 minute. The rate of reaction in terms of molarity per second is
 a) 1.06×10^{-6} b) 1.06×10^{-5} c) 1.06×10^{-7} d) 1.06×10^{-9}

More than One correct answer Type Questions

3. Which of the following statements is (are) true for the given reaction, $4A + B \rightleftharpoons 2C + 2D$?
 a) The rate of disappearance of B is one fourth the rate of disappearance of A
 b) The rate of appearance of C is half the rate of disappearance of B
 c) The rates of formation of C and D are equal
 d) The rate of formation of D is half the rate of disappearance of A
4. The rate equation for the decomposition of N_2O_5 in CCl_4 is, rate = $K[N_2O_5]$, where $K = 6.3 \times 10^{-4} s^{-1}$ at 320 K. What would be the initial rate of decomposition of N_2O_5 in a 0.10 M solution of N_2O_5 ?
 a) $6.3 \times 10^{-6} \text{ mol litre}^{-1} s^{-1}$ b) $0.63 \times 10^{-6} \text{ mol litre}^{-1} s^{-1}$
 c) $6.3 \times 10^{-5} \text{ mol litre}^{-1} s^{-1}$ d) $0.63 \times 10^{-4} \text{ mol litre}^{-1} s^{-1}$

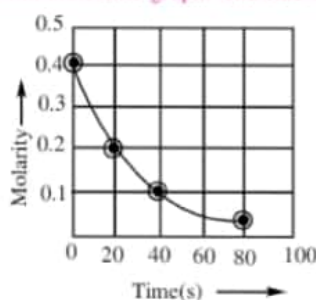
5. Select the correct statement(s).
- The rate of reaction decreases with decrease in temperature
 - The rate of reaction depends on nature of reactants
 - As time proceeds concentration of reactants decreases
 - As time proceeds concentration of products increases
6. SO_2 reacts with O_2 as follows $2\text{SO}_2 + \text{O}_2 \rightarrow 2\text{SO}_3$, the rate of disappearance of SO_2 is $2.4 \times 10^{-4} \text{ mole lit}^{-1} \text{ min}^{-1}$. Then
- Rate of reaction is $1.2 \times 10^{-4} \text{ mole lit}^{-1} \text{ min}^{-1}$
 - Rate of appearance of SO_3 is $2.4 \times 10^{-4} \text{ mole lit}^{-1} \text{ min}^{-1}$
 - Rate of disappearance of O_2 is $1.2 \times 10^{-4} \text{ mole lit}^{-1} \text{ min}^{-1}$
 - Rate of reaction is twice the rate of disappearance of SO_3

Linked Comprehension Type Questions

Passage-I:

For the reaction : $aA + bB \longrightarrow cC + dD$; Rate = $\frac{dx}{dt} = \frac{-1}{a} \frac{d[A]}{dt} = \frac{1}{b} \frac{d[B]}{dt} = \frac{1}{c} \frac{d[C]}{dt} = \frac{1}{d} \frac{d[D]}{dt}$

7. The rate of formation of SO_3 in the following reaction $2\text{SO}_2 + \text{O}_2 \longrightarrow 2\text{SO}_3$ is 100 g min^{-1} . Hence, the rate of disappearance of O_2 is
- 2 g min^{-1}
 - 20 g min^{-1}
 - 200 g min^{-1}
 - 50 g min^{-1}
8. A reaction follows the given concentration-time graph. The rate for this reaction at 20 s will be



- $4 \times 10^{-3} \text{ Ms}^{-1}$
 - $1 \times 10^{-2} \text{ Ms}^{-1}$
 - $2 \times 10^{-2} \text{ Ms}^{-1}$
 - $7 \times 10^{-3} \text{ Ms}^{-1}$
9. In the following reaction : $x\text{A} \longrightarrow y\text{B}$
- $$\log \left[-\frac{d[A]}{dt} \right] = \log \left[\frac{d[B]}{dt} \right] + 0.3$$
- Where negative sign indicates rate of disappearance of the reactant. Thus $x : y$ is
- 1 : 2
 - 2 : 1
 - 3 : 1
 - 3 : 10

Integer Type Questions

10. For a chemical reaction $aA \rightarrow bB$, $\log \left[-\frac{d[A]}{dt} \right] = \log \left[\frac{d[B]}{dt} \right] + 0.6$ then find the approximate ratio of a and b
11. In a gaseous phase reaction, $\text{A}_2(\text{g}) \rightarrow \text{B}(\text{g}) + (1/2)\text{C}(\text{g})$, the increase in pressure from 100 mm to 120 mm is noticed in 5 minute. The rate of disappearance of A_2 in mm min^{-1} is :

EXERCISE-II

(Order, molecularity and Half-life)

LEVEL-I (MAIN)

Straight Objective Type Questions

- For an elementary reaction, $2A + B \rightarrow C + D$, the active mass of B is kept constant but that of A is tripled. The rate of reaction will
 - decrease by 3 times
 - increase by 9 times
 - increase by 3 times
 - decrease by 6 times
- For a chemical reaction $Y_2 + 2Z \rightarrow \text{Product}$, rate controlling step is $Y + 1/2 Z \rightarrow Q$. If the concentration of Z is doubled, the rate of reaction will
 - Remain the same
 - Become four times
 - Become 1.414 times
 - Become double
- In a reaction $A \rightarrow B$, when the concentration of reactant is made 8 times, the rate got doubled. The order of reaction is
 - 1/3
 - 1
 - 1/2
 - 2
- The initial rates for gaseous reaction $A + 3B \rightarrow AB_3$ are given below

[A] (M)	[B] (M)	Rate (M sec ⁻¹)
0.1	0.1	0.002
0.2	0.1	0.002
0.3	0.2	0.008
0.4	0.3	0.018

Order of reaction is

- zero
 - three
 - one
 - two
- If the initial concentration is reduced to $1/4^{\text{th}}$ of the initial value of a zero order reaction, the half life of the reaction
 - remains constant
 - Becomes $1/4^{\text{th}}$
 - becomes double
 - Becomes fourfold
 - If $\frac{dx}{dt} = k[H_3O^+]^n$ and rate becomes 100 times when pH changes from 2 to 1. Hence order of reaction is
 - 1
 - 2
 - 3
 - 0
 - The initial concentration of cane sugar in presence of an acid was reduced from 0.20 to 0.10M in 5 hours and to 0.05M in 10 hours, value of K ? (in hr⁻¹)
 - 0.693
 - 1.386
 - 0.1386
 - 3.465
 - 50% completion of a first order reaction takes place in 16 minutes. Then fraction that would react in 32 minutes from the beginning
 - 1/2
 - 1/4
 - 1/8
 - 3/4

9. Consider the order of a reaction. Choose the incorrect statement(s).
- Order of a reaction may be zero, integer or fractional
 - Overall order of the reaction never be negative.
 - The order of an elementary step is always equal to its molecularity.
 - For the chemical equation $2\text{NH}_3 \xrightarrow{\Delta} \text{N}_2 + 3\text{H}_2$ the order of reaction is 4.
10. For a first order reaction, the half-life is 50 sec. Identify the correct statement from the following.
- the reaction is almost complete in 500 sec
 - the same quantity of reactant is consumed for every 50 sec of the reaction
 - quantity of reactant remaining after 100 sec is half of what remains after 50 sec
 - All the above three
11. For a first order reaction, if the time taken for completion of 50% of the reaction is 't' second, the time required for completion of 99.9% of the reaction is:
- 10t
 - 5t
 - 100t
 - 2t
12. In case of first order reactions
- $t_{3/4} = 2t_{1/2}$
 - $t_{7/8} = 3t_{1/2}$
 - $t_{99.9} = 10t_{1/2}$
 - All
13. The kinetic datas for the reaction: $2\text{A} + \text{B}_2 \rightarrow 2\text{AB}$ are as given below:
- | [A]
mol L ⁻¹ | [B ₂]
mol L ⁻¹ | Rate
mol L ⁻¹ min ⁻¹ |
|----------------------------|--|---|
| 2 | 1.0 | 2.5×10^{-3} |
| 1.0 | 1.0 | 5.0×10^{-3} |
| 2 | 2.0 | 1×10^{-2} |
- Hence the order of reaction w.r.t. A and B₂ are, respectively.
- 1 and 2
 - 2 and 1
 - 1 and 1
 - 2 and 2
14. The rate law of a reaction, $\text{A} + \text{B} \rightarrow \text{Product}$ is rate = $\text{K}[\text{A}]^n[\text{B}]^m$. On doubling the concentration of A and halving the concentration of B, the ratio of new rate to the earlier rate of reaction will be
- $n - m$
 - 2^{n-m}
 - $\frac{1}{2^{m+n}}$
 - $m + n$
15. Consider a reaction; $a\text{G} + b\text{H} \rightarrow \text{products}$. When concentration of both the reactants G and H is doubled, the rate increase by eight times. However, when concentration of G is doubled keeping the concentration of H fixed, the rate is doubled. The overall order of the reaction is;
- 0
 - 1
 - 2
 - 3
16. In a first order reaction, the concentration of reactant decreases from 800 mol/dm³ to 50 mol/dm³ in 2×10^4 sec. The rate constant in sec⁻¹ is: [log 2 = 0.3]
- 2×10^4
 - 3.45×10^{-5}
 - 1.386×10^{-4}
 - 2×10^{-4}
17. A reaction involving two different reactants can never be:
- Unimolecular reaction
 - I order reaction
 - II order reaction
 - Bi-molecular reaction

18. The rate equation for the reaction; $2A+B \rightarrow C$ is rate = $K[A][B]$. The correct statement about this is:
- 1) K is independent of $[A]$ and $[B]$
 - 2) $t_{1/2}$ with respect to A is dependent on initial concentration of A
 - 3) Unit of K is sec^{-1}
 - 4) Rate of formation of C is twice the rate of disappearance of A
19. The half life period of pyruvic acid in the presence of an amino transferase enzyme (which converts in to alanine) was found to be 221sec. How long will it take for the concentration of pyruvic acid to fall to $1/64$ of its initial value in this first order reaction?
- 1) 1200 sec
 - 2) 2440 sec
 - 3) 663 se
 - 4) 1326 sec

Numerical Value Type Questions

20. For the reaction $2A + B \rightarrow \text{Products}$, it is found that doubling the concentration of both reactants increases the rate by a factor of 8. But doubling the concentration of B alone, only doubles the rate. What is the order of the reaction w.r.t to A ?
21. Based on the following data for a reaction what is its order ($A \rightarrow \text{products}$)
- | | | | | |
|--------------|----|------|-------|----------|
| Conc. A | 2M | 0.2M | 0.02M | 0.00 |
| Time in min. | 0 | 10 | 20 | ∞ |
22. In a first order reaction, the concentration of the reactant, decreases from 0.8 M to 0.4 M in 15 minutes. The time taken for the concentration to change from 0.1 M to 0.025 M is ____ min.
23. Rate constant of a reaction with a virus is $3.3 \times 10^{-4} \text{ s}^{-1}$. Time (min) required for the virus to become 75% inactivated is: (Assume virus growth cannot take place simultaneously)

LEVEL-II (ADVANCED)

Straight Objective Type Questions

1. In a first order reaction, 20% reaction is completed in 24 minutes. The percentage of reactant remaining after 48 minutes is
- a) 60
 - b) 64
 - c) 81
 - d) 80
2. The concentration of the reactant A in the reaction $A \rightarrow B$ at different times are given below:

Concentration (M)	Time (seconds)
0.069	0
0.052	17
0.035	34
0.018	51

The rate constant of the reaction according to the correct order of reaction is

- a) $0.001 \text{ M}^{-2}\text{s}^{-1}$
 - b) $0.001 \text{ M}^{-2}\text{s}^{-1}$
 - c) 0.001 s^{-1}
 - d) 0.001 Ms^{-1}
3. For a first order reaction with half-life of 150 seconds, the time taken for the concentration of the reactant to fall from $M/10$ to $M/100$ will be approximately
- a) 1500 s
 - b) 500 s
 - c) 900 s
 - d) 600 s

4. For a first order reaction $A \rightarrow B$, the reaction rate at reactant concentration of 0.01 M is found to be $2.0 \times 10^{-5} \text{ mol L}^{-1} \text{ s}^{-1}$. The half life period of the reaction is
 a) 220s b) 30 s c) 374 s d) 347 s
5. For a first order reaction at 27°C , the ratio of time required for 75% completion to 25% completion of reaction is :
 a) 3.0 b) 2.303 c) 4.8 d) 0.477
6. Two first order reaction have half-lives in 1:2 ratio. After two half-lives of first reaction, What is the ratio of their rates of reactions ? (Initial concentrations same)
 a) 1 : 2 b) 1 : 3 c) 1 : 4 d) 1 : 1
7. A reaction $2A \longrightarrow 3B$ takes place according to first order kinetics the time in which the concentration of reactant and product become equal is ($K = 4.6 \times 10^{-2} \text{ min}^{-1}$, $\log 5 = 0.7$, $\log 3 = 0.48$, $\log 2 = 0.3$)
 a) 11 min b) 15 min c) 9 min d) 25 min
8. Heptene decomposes according to following reaction $\text{C}_7\text{H}_{14} \xrightarrow{\Delta} 2\text{C}_2\text{H}_4 + \text{C}_3\text{H}_6$. Rate constant was found to be $1 \times 10^{-4} \text{ sec}^{-1}$. In what time molar ratio of heptene to ethene in the reaction mixture will attain the value 1? [$\log 2 = 0.3010$, $\log 3 = 0.4771$]
 a) 68 min b) 4055 min c) 1736 min d) 228 min
9. The initial rate of a reaction if its rate constant is 10^{-3} min^{-1} when the initial concentration of reactant is 0.2 mol dm^{-3} and also percent amount of reactant converted into products in 200 minutes are respectively.
 a) $2 \times 10^{-4} \text{ mol dm}^{-3} \text{ min}^{-1}$, 81.97% b) $2 \times 10^{-4} \text{ mol dm}^{-3} \text{ min}^{-1}$, 18.03%
 c) $2 \times 10^{-4} \text{ mol dm}^{-3} \text{ min}^{-1}$, 76% d) $2 \times 10^{-4} \text{ mol dm}^{-3} \text{ min}^{-1}$, 24%

More than One correct answer Type Questions

10. Which of the following statements are correct?
 a) Time required for 75% completion is 1.5 times of half-life for zero order reaction.
 b) Time needed for a definite fraction of 1st order reaction does not vary with the initial concentration.
 c) Time for 25% reaction is one-third of half life in second order process.
 d) Rate of 1st order reaction gets doubled if the conc of the reactant is increased to a two fold value.
11. For a reaction, $A + B \rightarrow \text{products}$, the rate of reactions is increased by 8 times when the concentration of A & B are doubled. The rate of reaction is doubled when the concentration of A alone is doubled. Then
 a) Order of reaction with respect to A is 1 b) Order of reaction with respect to B is 2
 c) $\text{Rate} = K[A][B]^2$ d) Units of rate constant are $\text{mole}^{-2} \text{ lit}^2 \text{ time}^{-1}$
12. For a reaction $A \rightarrow \text{products}$, the rate of reaction is increased by 8 times when the concentration of A is doubled then
 a) order of the reaction is 4 b) order of the reaction is 3
 c) units of rate constant are $\text{mole}^{-2} \text{ lit}^2 \text{ time}^{-1}$ d) $\text{rate} = K[A]^3$
13. In case of zero order reactions
 a) $t_{1/2} = 2t_{1/4}$ b) $t_{3/4} = 3t_{1/2}$ c) $t_{\infty} = 2t_{1/2}$ d) $t_{90\%} = 3t_{50\%}$

14. Identify incorrect statements

- a) Molecularity is not always an integral quantity
- b) Molecularity can be defined for elementary reactions only
- c) Order of reaction does not exceed three
- d) Molecularity became zero

15. Hydrolysis of an ester is catalysed by H^+ ion. Using equimolar concentrations of two acids HX and HY both being strong acids. The rate constants of the reaction are found to be $3 \times 10^{-3} \text{ min}^{-1}$ & $5 \times 10^{-3} \text{ min}^{-1}$ respectively at a fixed temperature. It can be concluded that.

- a) Rate const may be taken as the measure of degree of ionization of one acid used as catalyst.
- b) HX is a stronger acid than HY, their relative strength being 1.7
- c) HX is a weaker acid than HY.
- d) HX is a stronger base than HY.

16. $Zn + 2H^+ \rightarrow Zn^{2+} + H_2$

Half-life period is independent of concentration of zinc at constant pH. For the constant concentration of Zn, rate becomes 100 times when pH is decreased from 3 to 2. Hence

a) $\frac{dx}{dt} = k[Zn]^0[H^+]^2$

b) $\left(\frac{dx}{dt}\right) = k[Zn][H^+]^2$

- c) rate is not affected if concentration of zinc is made four times and that of H^+ ion is halved.
- d) rate becomes four times if concentration of H^+ ion is doubled at constant Zn concentration.

17. The rate law for the reaction : $RCI + NaOH \rightarrow ROH + NaCl$. Is given by : rate = $k [RCI]$. The rate of this reaction

- a) is doubled by doubling the concentration of NaOH
- b) is halved by reducing the concentration of RCI by one half
- c) is increased by increasing the temperature of the reaction
- d) is unaffected by change in temperature

Linked Comprehension Type Questions

Passage-I :

Consider the reaction represented by the equation : $CH_3Cl(g) + H_2O(g) \rightarrow CH_3OH(g) + HCl(g)$
These kinetic data were obtained for the given reaction concentrations :

Initial conc (M)		Initial rate of disappearance of $CH_3Cl(Ms^{-1})$
$[CH_3Cl]$	$[H_2O]$	
0.2	0.2	1
0.4	0.2	2
0.4	0.4	8

18. The rate law for the reaction will be

a) $r = k [CH_3Cl][H_2O]$

b) $r = k [CH_3Cl]^2[H_2O]$

c) $r = k [CH_3Cl] [H_2O]^2$

d) $r = k [CH_3Cl]^2[H_2O]^4$

19. Overall order of the reaction will be

- a) 0 b) 1 c) 2 d) 3

20. If H_2O is taken in large excess, the order of the reaction will be

- a) 1 b) 0 c) 3 d) 2

Passage-II :

The following data were observed for the following reaction at 25°C .



Set	Initial concentration (M)		time (t) (min)	Final concentration (M)
	$[\text{A}]_0$	$[\text{B}]_0$		
I	0.10	0.05	25	0.0033
II	0.10	0.10	15	0.0039
III	0.20	0.10	7.5	0.0077

21. Rates $\frac{d[\text{C}]}{dt}$ in sets I, II and III are respectively, (in M min^{-1}) :

- | I | II | III |
|--------------------------|-----------------------|-----------------------|
| a) 1.30×10^{-4} | 2.6×10^{-4} | 1.02×10^{-3} |
| b) 0.033 | 0.0039 | 0.0077 |
| c) 0.02×10^{-4} | 0.04×10^{-4} | 0.017 |
| d) None of above | | |

22. Rate constants of the above experiment is (in $\text{L}^2\text{M}^{-2} \text{min}^{-1}$)

- a) 2.6×10^{-2} b) 2.6×10^{-1} c) 2.6×10^{-4} d) 1.3×10^{-2}

23. Rate law of the above experiment is

- a) $k[\text{A}]^2[\text{B}]$ b) $k[\text{A}][\text{B}]$ c) $k[\text{A}][\text{B}]^2$ d) $k[\text{A}]^2[\text{B}]^2$

Matrix Matching Type Questions

24. **Column-I**

- A) Molecularity of a reaction
B) Order of a reaction
C) The dissociation of H_2O_2 (aq)
D) $\text{H}_{2(\text{g})} + \text{Cl}_{2(\text{g})} \xrightarrow{\text{h}\nu} 2\text{HCl}$

Column-II

- p) 0,1 possible
q) 1,2 possible
r) First order
s) Zero order
t) Always only +ve integer
u) $t_{1/2}$ independent on initial concentration

25. **Column-I (Half-life)**

- A) $t_{1/2} = \text{constant}$
B) $t_{1/2} \propto a$
C) $t_{1/2} \propto p^{-1}$
D) $t_{1/2} \propto 1/a$

Column-II (Order)

- p) First order
q) Pseudo first order
r) Zero order
s) Second order

26. Column-I (Reaction)

- A) $\text{SO}_2\text{Cl}_2 \rightarrow \text{SO}_2 + \text{Cl}_2$
 B) $\text{CH}_3\text{COOC}_2\text{H}_5 + \text{NaOH} \rightarrow \text{CH}_3\text{COONa} + \text{C}_2\text{H}_5\text{OH}$
 C) $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$
 D) $\text{H}_2\text{O}_2 + 2\text{I}^- + 2\text{H}^+ \rightarrow 2\text{H}_2\text{O} + \text{I}_2$

Column-II (Units of K)

- p) S^{-1}
 q) min^{-1}
 r) $\text{L.mole}^{-1} \text{ min}^{-1}$
 s) $\text{L.mole}^{-1} \text{ S}^{-1}$

27. Column-I

- A) Decomposition of SO_2Cl_2
 B) $\text{RCOOR} + \text{NaOH} \rightarrow \text{RCOONa} + \text{ROH}$
 C) Decomposition of N_2O_5
 D) $\text{RCOOR} + \text{H}_2\text{O} \xrightarrow{\text{H}^+} \text{RCOOH} + \text{ROH}$

Column-II

- p) First order reaction
 q) Units of rate constant sec^{-1}
 r) Half life is independent of initial conc
 s) Pseudo unimolecular reaction
 t) Units of rate of reaction are $\text{M}^{-1} \cdot \text{time}^{-1}$

28. Column-I

Linear plots (with non zero slope)

- A) $\ln \left[\frac{d[A]}{dt} \right]$ vs. $\ln[A]$
 B) $\log k$ vs. $\frac{1}{T}$
 C) $\log t_{1/2}$ vs. $\log[A]_0$
 D) $\frac{-d[A]}{dt}$ vs. $[A]^2$

Column-II

(Order)

- p) 2
 q) $\frac{1}{2}$
 r) 0
 s) 1

Integer Type Questions

29. The concentration of R in the reaction $\text{R} \rightarrow \text{P}$ was measured as a function of time and the following data is obtained :

[R] (molar)	1.0	0.75	0.40	0.10
t(min.)	0.0	0.05	0.12	0.18

The order of the reaction is

30. Half-life period of the decomposition of a compound is 20 minutes. If the initial concentration is doubled, the half-life period is reduced to 10 mins. What is the order of the reaction?
31. For the first order reaction $t_{99.99\%} = xt_{90\%}$ then, find the value of x .
32. For a reaction, the graph drawn between $t_{1/2}$ and 'a' gives a straight line passing through the origin with the slope $2 \times 10^2 \text{ mole}^{-1} \cdot \text{lit} \cdot \text{min}$. If the initial concentration of the reactant is 1M, then the half life period is $____ \times 10^2 \text{ min}$.
33. For hypothetical chemical reaction $\text{A} \rightarrow \text{I}$, it is found that the reaction is third order in A. What happens to the rate of reaction when the concentration of A is doubled rate increase by x times. Then x value is
34. The conversion of molecules of A to B follows second order kinetics. Doubling the concentration of 'A' will increase the rate of formation of 'B' by a factor of ?

EXERCISE-III

(Theories of Chemical Kinetics)

LEVEL-I (MAIN)

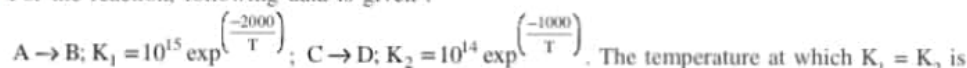
Straight Objective Type Questions

- For the reaction $A + B \rightleftharpoons C + D$, the forward reaction is exothermic. The activation energy of formation of $A + B$ is ____ that for the formation of $C + D$
 - equal to
 - less than
 - greater than
 - double
- The minimum energy required for molecules to enter into chemical reaction is called
 - Kinetic energy
 - Potential energy
 - Threshold energy
 - Activation energy
- Wrong statement among the following is
 - effective collisions are more if activation energy is less
 - zero order reaction proceeds at a constant rate independent of concentration or time
 - reactions with highest rate constant values have lowest activation energies
 - if initial concentration increases half life decreases in zero order
- The population of activated molecules can be increased by
 - increase in temperature
 - using a catalyst
 - increase of concentration of reactants
 - All
- Collision theory satisfactorily explains
 - First order reaction
 - Zero order reaction
 - Bimolecular reaction
 - Any order reaction
- An endothermic reaction $A \rightarrow B$ has an activation energy as $x \text{ kJ.mol}^{-1}$ of A . If energy change of the reaction is $y \text{ kJ}$, the activation energy of the reverse reaction is
 - $-x$
 - $x - y$
 - $x + y$
 - $y - x$
- Which of the following explains the increase of reaction rate by a catalyst ?
 - Catalyst provides the necessary energy to the colliding molecules to cross the barrier
 - Catalyst decreases the rate of backward reaction so that the rate of forward reaction increases
 - Catalyst decreases the enthalpy change of the reaction
 - Catalyst provides an alternative path of lower activation energy
- In a reaction, threshold energy is equal to
 - activation energy
 - normal energy of the reactants
 - activation energy + energy of reactants
 - activation energy – energy of reactants
- For reaction $A \rightarrow B$, the rate constant $k_1 = A_1 e^{-E_{a1}/RT}$ and for the reaction $P \rightarrow Q$, the rate constant $k_2 = A_2 e^{-E_{a2}/RT}$. If $A_1 = 10^8$, $A_2 = 10^{10}$ and $E_{a1} = 600$, $E_{a2} = 1200$, then the temperature at which $k_1 = k_2$ is
 - $\frac{600}{R} \text{ K}$
 - $\frac{300 \times 4.606}{R} \text{ K}$
 - $\frac{600}{4.606R} \text{ K}$
 - $\frac{4.606}{600R} \text{ K}$

10. If a reaction $A + B \rightarrow C$ is exothermic to the extent of 30 KJ/mol and the forward reaction has an activation energy of 70 KJ/mol, the activation energy for the reverse reaction is
 1) 30 KJ/mol 2) 40KJ/mol 3) 70KJ/mol 4) 100KJ/mol
11. Unit of frequency factor A in : $K = Ae^{-E_a/RT}$ is ;
 1) time^{-1} 2) $\text{mole litre}^{-1}\text{time}^{-1}$
 3) $\text{litre mol}^{-1}\text{t}^{-1}$ 4) dependent of order of reaction
12. For the decomposition of $N_2O_5(g)$, it is given that: $2N_2O_5(g) \rightarrow 4NO_2(g) + O_2(g)$. Activation energy E_a , $N_2O_5(g) \rightarrow 2NO_2(g) + \frac{1}{2}O_2(g)$; Activating energy E'_a ,
 1) $E_a = E'_a$ 2) $E_a > E'_a$ 3) $E_a < E'_a$ 4) $E_a = 2E'_a$
13. Effective collision are those in which molecules must:
 1) Have energy equal to or greater than the threshold energy
 2) Have proper orientation
 3) Acquire the energy of activation
 4) All of these
14. The activation energy of a reaction is zero. The rate constant of the reaction:
 1) Increases with increase of temperature 2) Decreases with decrease of temperature
 3) Decreases with increase of temperature 4) Independent of temperature
15. Which reaction characteristics are changing by the addition of a catalyst to a reaction at constant temperature
 1) Activation energy 2) Equilibrium constant
 3) Reaction entropy 4) Reaction enthalpy
16. For a first order reaction $A \rightarrow P$, the temperature (T) dependent rate constant (K) was found to follow the equation $\log k = -(2000)\frac{1}{T} + 6.0$. The pre-exponential factor A and the activation energy E_a , respectively, are
 1) $1 \times 10^6 \text{s}^{-1}$ & 9.2KJ mol^{-1} 2) 6s^{-1} & 16.6KJ mol^{-1}
 3) $1 \times 10^6 \text{s}^{-1}$ & 16.6 KJ mol^{-1} 4) $1 \times 10^6 \text{s}^{-1}$ & 38.3 KJ mol^{-1}
17. The activation energy of a reaction is 58.3 kJ/mole. The ratio of the rate constants at 305K and 300K is about ($R = 8.3\text{Jk}^{-1}\text{mol}^{-1}$) (Antilog 0.1667 = 1.468)
 1) 1.25 2) 1.75 3) 1.5 4) 2.0
18. For $X \rightarrow Y$, $\frac{k_{t+10}}{k_t} = 3$. If the rate constant at 300 k is 'Q' min^{-1} , at what temperature rate constant becomes '9Q' min^{-1} ?
 1) 47°C 2) 320°C 3) 280 K 4) $\sqrt{9 \times 300}$ K
19. For $N_2O_5(\text{in } CCl_4) \rightarrow 2NO_2 + \frac{1}{2}O_2$, $K = 6 \times 10^{-4} \text{s}^{-1}$ at 350K and $K = 1.2 \times 10^{-3} \text{s}^{-1}$ at 360 K. Then, when temperature is changed to 380 K, value of K (in s^{-1})
 1) 1.2×10^{-3} 2) 2.4×10^{-3} 3) 4.8×10^{-4} 4) 4.8×10^{-3}

Numerical Value Type Questions

20. For the reaction, following data is given :



21. At 52°C the activation energy $E_{a(b)}$ for a reaction is 2×10^{-2} k.cal and enthalpy of reaction at same temperature is 0.12 k.cal. Calculate % of reactant molecules which crossover the energy barrier.
22. The rate of a chemical reaction doubles for every 10°C rise in temperature. If the temperature is increased by 60°C, the rate of reaction increases by about ____ times.

LEVEL-II (ADVANCED)

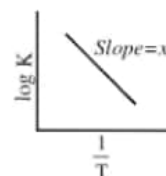
Straight Objective Type Questions

- Activation energy depends on
 - pressure
 - concentration of reactants
 - concentration of products
 - nature of reactants
- The rate constant K_1 of a reaction is found to be double that of rate constant K_2 of another reaction. The relationship between corresponding activation energies of the two reactions E_1 and E_2 can be represented as
 - $E_1 > E_2$
 - $E_1 < E_2$
 - $E_1 = E_2$
 - $E_1 = 2E_2$
- In the Arrhenius equation, the Boltzmann factor $e^{-E_a/RT}$ represents the ____ of the molecules possessing energy in excess of activation energy
 - number
 - fraction
 - weight
 - percentage
- Rate constant of a reaction can be expressed by Arrhenius equation as $K = Ae^{-E_a/RT}$. In this equation, E represents
 - the fraction of molecules with energy greater than the activation energy of the reaction
 - the energy above which all the colliding molecules will react
 - the energy below which colliding molecules will not react
 - the total energy of the reacting molecules at a temperature, T
- Activation energies for different reactions are given below.
 - $A \rightarrow \text{products}, E_a = 14 \text{ K.Cal}$
 - $B \rightarrow \text{products}, E_a = 16 \text{ K.cal}$
 - $C \rightarrow \text{products}, E_a = 12 \text{ K.Cal}$
 - $D \rightarrow \text{products}, E_a = 10 \text{ K.cal}$

If the temperature increases by 10°C for which reactions the temperature coefficients $\left(\frac{k_{t+10}}{k_t}\right)$ are maximum and minimum respectively.

- a & b
 - b & c
 - b & d
 - d & b
6. The graph of $\log K$ versus $1/T$ is given below. Here 'x' is equal to [Where K = rate constant and T = absolute temperature]

- $\frac{-E_a}{2.303}$
- $\frac{-E_a}{2.303R}$
- $\frac{-2.303}{E_a \cdot R}$
- $\frac{-2.303R}{E_a}$



7. The effect of temperature on a reaction rate for which E_a is zero is given by
 a) with increase of temperature rate increases b) with increase of temperature rate decreases
 c) rate is independent of temperature d) reaction never occurs
8. A first order reaction is 50% completed in 20 minutes at 27°C and in 5 minutes at 47°C . The energy of activation of the reaction is :
 a) 43.85 kJ/mol b) 55.14 kJ/mol c) 11.97 kJ/mol d) 6.65 kJ/mol
9. A first order reaction is 12.5% completed in 20 minutes at 300 K. When same reaction is carried out at 315 K, it is 12.5% complete in 2.5 minute. What is the activation energy of reaction (in k cal/mol)? [Given: $R = 2$ cal/mol and $\ln 2 = 0.7$]
 a) 35.38 b) 26.46 c) 17.69 d) 26.56

More than One correct answer Type Questions

10. Arrhenius equation may be represented as
 a) $\ln \frac{A}{k} = \frac{E_a}{RT}$ b) $\frac{d \ln k}{dT} = \frac{E_a}{RT}$
 c) $\log A = \log k + \frac{E_a}{2.303 RT}$ d) $\log \left(-\frac{E_a}{RT} \right) = \frac{k}{A}$
11. Which of the following is correct for Arrhenius equation $k = Ae^{-E_a/RT}$
 a) k is the rate of reaction at zero concentration of reactants
 b) A may be termed as the rate constant at very high temperature
 c) A may be termed as the rate constant at zero activation energy
 d) E_a is the activation energy of reactants
12. Which of the following statements are correct about the Arrhenius equation?
 a) The Arrhenius constant becomes equal to rate constant at a very high temperature
 b) When E_a is zero, rate is independent of temperature
 c) $e^{-E_a/RT}$ represents the fraction of molecules having energy more than threshold energy
 d) Positive catalyst increases E_a of the reaction
13. Pick out the correct Statements.
 a) Activated complex is an unstable intermediate with maximum energy and normal bonds
 b) Activated complex is an unstable intermediate with maximum energy and fractional bond orders
 c) Transition state cannot exist independently, it has greater vibrational character than unstable intermediate
 d) Transition state is a state in which bonds are half broken and half formed
14. In Arrhenius equation
 a) The Arrhenius constant has units of rate of reaction
 b) The Arrhenius constant has the units of rate constant of reaction
 c) If E_a is high, rate of reaction is slow
 d) If E_a is high, rate of reaction is high

Linked Comprehension Type Questions

Passage-I :

For a chemical reaction, $A \rightarrow \text{products}$, the following equation is found to be followed,

$$\log K = 16.398 - \frac{2800}{T}$$

15. Arrhenius factor for the reaction is ____
 a) 2.5×10^{16} b) 5×10^{16} c) 7.5×10^{16} d) 4×10^{16}
16. Activation energy of the reaction is ____ K.Cal
 a) 128.13 b) 12.767 c) 12813 d) 1.2813
17. At 27°C the rate constant of the reaction is ____
 a) 1.16×10^5 b) 1.16×10^6 c) 1.321×10^7 d) 1.16×10^7

Passage-II :

Arrhenius studied the effect of temperature on the rate of a reaction and postulated that rate constant varies with temperature exponentially as $k = Ae^{-E_a/RT}$. For most of the reactions, it was found that the temperature coefficient of the reaction lies between 2 to 3. The method is generally used for finding the activation energy of a reaction. Keeping temperature constant, the effect of catalyst on the activation energy has also been studied by studying how much the rate of reaction changes in the presence of catalyst. In most of the cases, it is observed that catalyst lowers the activation energy barrier and increases the rate of reaction.

18. The pre-exponential factor in the Arrhenius equation of a second order reaction has the units
 a) $\text{mol L}^{-1} \text{s}^{-1}$ b) $\text{L mol}^{-1} \text{s}^{-1}$ c) s^{-1} d) dimensionless
19. Which of the following plot will be linear?
 a) $\ln k$ versus T with -ve slope b) k versus $1/T$ with -ve slope
 c) $\ln k$ versus $1/T$ with -ve slope d) $\ln k$ versus $1/T$ with +ve slope
20. If the rate of reaction grows 15.6 times on increasing the temperature by 30 K the temperature coefficient of the reaction will be nearly
 a) 2 b) 2.5 c) 3.0 d) 3.5

Passage-III :

A certain endothermic reaction : $A \rightarrow \text{Product}$ $\Delta H = +ve$ proceeds in a sequence of three elementary steps with the rate constants K_1 , K_2 and K_3 and each one having energy of activation E_1 , E_2 and E_3 respectively at 25°C . The observed rate constant for the reaction is equal to $K_3 \sqrt{\frac{K_1}{K_2}}$. A_1 , A_2 and A_3 are Arrhenius parameters respectively.

21. The observed energy of activation for the reaction is
 a) $\frac{2E_1 - E_2 + 2E_3}{2}$ b) $\frac{E_2 - E_1 - 2E_3}{2}$ c) $\sqrt{\frac{E_1 E_3}{E_2}}$ d) $\frac{E_1 - E_2}{2} + E_3$
22. The observed Arrhenius parameter for the reaction is
 a) $\frac{2A_1 - A_2 + 2A_3}{2}$ b) $\sqrt{\frac{A_1}{A_2}} \cdot A_3$ c) $A_1 A_2 A_3$ d) $\frac{A_1 \cdot A_3}{A_2}$

23. Presence of a catalyst decreases the energy of activation of each path by half the value of E_1 . Assuming the other factors same, the observed energy of activation would be

- a) $E_3 - \frac{E_2}{2}$ b) $2E_3 - E_2$ c) $E_2 - \frac{E_3}{2}$ d) $\frac{E_1}{2} + \frac{E_2}{2} + E_3$

Matrix Matching Type Questions

24. Column-I

A) Decomposition of H_2O_2

B) $\frac{k_{308K}}{k_{298K}}$

C) Arrhenius equation

D) $t_{99.9\%}$

Column-II

p) Ten half-lives

q) 1st order

r) Temperature coefficient

s) $\log \frac{k_2}{k_1} = \frac{E_a}{2.303R} \left(\frac{T_2 - T_1}{T_1 T_2} \right)$

t) 2 to 3

25. Column-I

A) $e^{-E_a/RT}$

B) A

C) $\frac{-k}{2.303}$

D) $t_{1/2} = \frac{a}{2k}$

Column-II

p) $\tan \theta, \theta > \frac{\pi}{2}$

q) Boltzmann's factor

r) Zero order reaction

s) Arrhenius factor

Integer Type Questions

26. The energy of activation for a reaction is 100 kJ/mol at 57°C. Presence of catalyst lower activation energy by 25%. The rate of reaction is increased by 10^x , the value of x is
27. The rate of reaction becomes two times for every 10°C rise in temperature. If the rate of reaction increases by 32 times when the temperature is increased from 30°C to $(10x)^\circ\text{C}$. Then $x =$
28. A first order chemical reaction, $A \rightarrow P$ is occurring at 500 K. If the same reaction is to be carried out in presence of a catalyst, temperature of 400 K is required to be maintained in order to maintain the same rate of reaction. If the catalyst lowers the activation energy by 30 kJ, then what would

EXERCISE-IV

(1st order, Parallel, Sequential, Eqm reactions, reaction mechanism)

LEVEL-I (MAIN)

Straight Objective Type Questions

1. In the reversible reaction : $2NO_2 \xrightleftharpoons[k_2]{k_1} N_2O_4$ the rate of disappearance of NO_2 is equal to

1) $\frac{2k_1}{k_2}[NO_2]^2$

2) $2k_1[NO_2]^2 - 2k_2[N_2O_4]$

3) $2k_1[NO_2]^2 - k_2[N_2O_4]$

4) $(2k_1 - k_2)[NO_2]$

2. Benzene diazonium chloride (A) decomposes into chlorobenzene (B) and $N_2(g)$ in first-order reaction. Volumes of N_2 collected after 5 min and at the complete decomposition of A are 10 mL and 50 mL respectively. The rate constant for the reaction is
 1) 0.446 min^{-1} 2) 0.0446 min^{-1} 3) 0.223 min^{-1} 4) 0.112 min^{-1}
3. For a reaction $A_{(g)} \rightarrow B_{(g)} + C_{(g)} + D_{(g)}$ at 750K the rate constant is $2.3 \times 10^{-3} \text{ min}^{-1}$. Starting with a pressure of 400 mm of Hg at this temperature in a closed container, how many minutes would it take for the pressure in the container to become 760 mm Hg. ($\log 11 = 1.041$)
 1) 360 2) 460 3) 260 4) 160
4. Consider the following reactions $A + B \xrightleftharpoons[k_{-1}]{k_1} C$, $C + B \xrightarrow{k_2} D$. If $\frac{d[C]}{dt} = 0$ (i.e., steady state). Then $\frac{-d[A]}{dt}$ is equal to
 1) $\frac{k_1[A][B]}{k_{-1} + k_2[B]}$ 2) $\frac{k_1 k_2 [A][B]^2}{k_{-1} + k_2[B]}$ 3) $\frac{2k_1 k_2 [A][B]}{k_{-1} + k_2[B]}$ 4) $\frac{k_1 k_2 [A]^2 [B]}{k_{-1} + k_2[B]}$
5. The inversion of cane sugar proceeds with half-life of 500 minutes at $P_H = 5$ for any concentration of sugar. However if $P_H = 6$; the half life changed to a 50 minutes. The rate law expression for sugar inversion can be written as
 1) $r = k[\text{sugar}]^2[H^+]^0$ 2) $r = k[\text{sugar}]^1[H^+]^0$
 3) $r = k[\text{sugar}]^1[H^+]^1$ 4) $r = k[\text{sugar}]^0[H^+]^1$

Numerical Value Type Questions

6. A reactant 'A' decomposes to produce two different products simultaneously, but at different rate as shown below:



The order of reaction is unity for both reaction and their respective rate constants K_1 & K_2 are 1.2×10^{-2} & 3×10^{-2} per second, respectively. The activation energy for overall reaction (KJ/mol) (if the activation energy for individual reactions E_1 and E_2 are 180 & 200 KJ/mol respectively) is

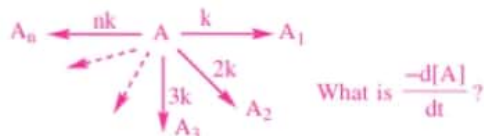
LEVEL-II (ADVANCED)

Single Correct Answer type Questions

1. Which integrated equation is correct for the reaction given below? $A(g) \rightarrow B(g) + C(g) + D(g)$
 $P_i \rightarrow$ initial pressure ; $P_t \rightarrow$ total pressure
 a) $k = \frac{2.302}{t} \log_{10} \left[\frac{P_i}{P_t} \right]$ b) $k = \frac{2.302}{t} \log_{10} \left[\frac{P_i}{P_t} \right]$
 c) $k = \frac{2.302}{t} \log_{10} \left[\frac{2P_t}{3P - P_t} \right]$ d) $k = \frac{2.302}{t} \log_{10} \left[\frac{3P_t}{2P - 3P_t} \right]$
2. For the formation of phosgene from CO(g) and chlorine (g) as, $CO(g) + Cl_2(g) \rightarrow COCl_2(g)$ the possible mechanism is $Cl_2 \rightleftharpoons 2Cl$ (fast); $Cl + CO \rightleftharpoons COCl$ (fast); $COCl + Cl_2 \rightarrow COCl_2 + Cl$ (slow). The rate law expression can be written as
 a) Rate = $k [CO] [Cl_2]$ b) Rate = $k [CO]^{1/2} [Cl_2]$
 c) Rate = $k [CO] [Cl_2]^{3/2}$ d) Rate = $k [CO] [Cl_2]^{1/2}$

3. For any elementary reaction $2A \xrightleftharpoons[k_2]{k_1} B$, the rate of disappearance of A is equal to
- a) $\frac{2k_1}{k_2}[A]^2$ b) $-2k_1[A]^2 + 2k_2[B]$ c) $2k_1[A]^2 - 2k_2[B]$ d) $(2k_1 - k_2)[A]$

4. Mechanism of the reaction is :



- a) $\frac{n(n+1)}{2}k[A]$ b) $k[A]$ c) $nk[A]$ d) $\frac{2}{n(n+1)}k[A]$
5. Mechanism of the reaction is $A_2 \xrightleftharpoons[k_{-1}]{k_1} 2A$; $A + B \xrightarrow{k_2} C$; $A_2 + C \xrightarrow{k} D + A$. What is $-\frac{d[A_2]}{dt}$?
- a) $k_1[A_2] - k_{-1}[A]^2 + k[A_2][C]$ b) $2k_1[A]^2 - k_1[A_2] + k_2[A][B] - k[A_2][C]$
- c) $k_2[A][B] - k[A_2][C]$ d) $1[A_2][C]$

Linked Comprehension Type Questions

Passage-I :

Consider the reaction $2A \longrightarrow 4B + C$;

optically active A, decomposes into B and C, which are also optically active, if rotation per mole of A is 10° , B is 30° and C is -40° respectively. After 10 sec the rotation of mixture was 25° and at infinite time was 30° . [Degree of dissociation is fraction dissociated per mole]

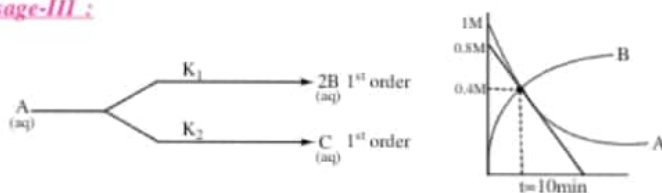
6. Rate constant of reaction is
- a) $\frac{1}{10} \ln \frac{9}{2}$ b) $\frac{1}{10} \ln \frac{3}{2}$ c) $\frac{1}{10} \ln \frac{4}{9}$ d) $\frac{1}{10} \ln \frac{2}{9}$
7. Degree of dissociation of A at 10 sec is
- a) $\alpha = \frac{7}{9}$ b) $\alpha = \frac{3}{4}$ c) $\alpha = \frac{7}{12}$ d) $\alpha = \frac{12}{9}$
8. Degree of dissociation of A at 20 sec is
- a) $\alpha = \frac{77}{81}$ b) $\alpha = \frac{81}{77}$ c) $\alpha = \frac{3}{4}$ d) $\alpha = \frac{6}{13}$

Passage-II :

Elementary unimolecular reactions have first order rate laws, elementary bimolecular reactions have second order rate laws. A rate law is often derived from a proposed mechanism by imposing the steady state approximation or by assuming that there is a pre-equilibrium. A proposed mechanism must be consistent with the experimental rate law.

9. The molecularity of the elementary reaction $C_2N_2 \rightarrow 2CN$ is :
- a) zero b) one c) two d) three
10. In a gas-phase reaction, a reaction takes place only if the reactant molecules get ____ during the molecular collisions give products.
- a) threshold energy b) activation energy c) both a & b d) heat of reaction

Passage-III:



11. What is the half life of the given reaction?
 a) 8min b) 7.5min c) 3.5min d) 10min
12. What is the ratio of the rate constants K_1 and K_2 ?
 a) 2 : 3 b) 1 : 3 c) 2 : 5 d) 1 : 2
13. What is the ratio of concentrations of 'B', 'C' at any stage of the reaction?
 a) 1 b) 2 c) 3 d) 4

Integer Type Questions

14. A tiny spherical ball of water soluble solute is dropped in 1 litre of water. Mass of ball is 1/7 gm and mol mass of solute is 125. Decrease in surface area due to dissolution follows 1st order of half life period 1 sec. and after 2 sec. molarity of solution is 10^{-x} . Find the value of x.
15. A chemical reaction occurs in three paths having rate constants k_1 , k_2 & k_3 respectively. If E_{a1} , E_{a2} and E_{a3} are 4, 5 & 8kJ respectively and overall rate constant $k = \frac{k_1 k_3}{k_2}$. Assuming $A_{av} = \frac{A_1 A_3}{A_2}$, the overall energy of activation in kJ is ____

KEY SHEET (LECTURE SHEET)

EXERCISE-I

LEVEL-I

- 1) 2 2) 3 3) 2 4) 1 5) 1 6) 4 7) 3 8) 4
 9) 3 10) 1.6 11) 0.1 12) 2

LEVEL-II

- 1) a 2) b 3) acd 4) cd 5) abcd 6) abc 7) b 8) d
 9) b 10) 4 11) 8

EXERCISE-II

LEVEL-I

- 1) 2 2) 3 3) 1 4) 1 5) 2 6) 2 7) 3 8) 4
 9) 4 10) 3 11) 1 12) 4 13) 1 14) 2 15) 4 16) 3
 17) 1 18) 1 19) 4 20) 2 21) 0 22) 30 23) 70

LEVEL-II

- 1) b 2) d 3) b 4) d 5) c 6) d 7) a 8) a
 9) b 10) abcd 11) abcd 12) bcd 13) ac 14) ac 15) ac 16) bcd
 17) bcd 18) c 19) d 20) a 21) a 22) b 23) a
 24) A-qt; B-pq; C-ru; D-s 25) A-pq; B-r; C-s; D-s
 26) A-pq; B-rs; C-pq; D-rs 27) A-pq; B-t; C-r; D-s
 28) A-pqs; B-pqrs; C-pqr; D-p 29) 0 30) 2 31) 4
 32) 2 33) 8 34) 4

EXERCISE-III

LEVEL-I

- 1) 3 2) 4 3) 4 4) 4 5) 3 6) 2 7) 4 8) 3
 9) 3 10) 4 11) 4 12) 1 13) 4 14) 4 15) 1 16) 4
 17) 3 18) 1 19) 4 20) 434.22 21) 80.6 22) 64

LEVEL-II

- 1) d 2) b 3) b 4) c 5) c 6) b 7) c 8) b
 9) b 10) ac 11) bcd 12) abc 13) bd 14) bc 15) a 16) b
 17) d 18) b 19) c 20) b 21) d 22) b 23) a
 24) A-q; B-rst; C-rs; D-p 25) A-q; B-s; C-p; D-r 26) 4 27) 8
 28) 6

EXERCISE-IV

LEVEL-I

- 1) 2 2) 2 3) 3 4) 2 5) 2 6) 194.7

LEVEL-II

- 1) c 2) c 3) c 4) a 5) a 6) a 7) a 8) a
 9) b 10) b 11) b 12) d 13) a 14) 3 15) 7

PRACTICE SHEET

EXERCISE-I

(Introduction, rate of reactions Factors influencing rate, specific rate)

LEVEL-I (MAIN)

Straight Objective Type Questions

- Which of the following reactions occurs at measurable rate?
 - reaction between H^+ and OH^- ions in aqueous solution
 - reaction between $AgNO_3$ and $NaCl$ aqueous solutions
 - hydrolysis of methyl acetate
 - reaction between hydrogen and oxygen gases at room temperature
- Which of the following reaction is spontaneous at room temperature?
 - $I_2 + H_2O \rightarrow HI + HIO$
 - $2H_2 + O_2 \xrightarrow{Pt} 2H_2O$
 - $N_2 + O_2 \rightarrow 2NO$
 - $2HCl \rightarrow H_2 + Cl_2$
- The rate of a chemical reaction
 - increases as the reaction proceeds
 - decreases as the reaction proceeds
 - may increase or decrease during the reaction
 - remains constant as the reaction proceeds
- The chemical reaction occurring between covalent molecules involves
 - breaking of existing bonds
 - formation of new bonds
 - evolution of heat energy
 - 1 & 2